

Occupational AI Exposure: Predicting Job Displacement Risk from O*NET Skill Profiles

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This project builds predictive models estimating AI-driven job displacement risk for 617 unique U.S. occupations, using measurable skill, ability, and knowledge features from the O*NET database as predictors. The project identifies which specific occupational characteristics most strongly predict automation risk and visualizes how that risk is distributed across job types and U.S. states.

Project Structure

FinalProjDv/

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└─ Data Files

| — automation_data_by_state.csv # Frey & Osborne automation scores + state employment

| — Skills.txt # O*NET skill ratings (tab-separated)

| — Abilities.txt # O*NET ability ratings (tab-separated)

| | Knowledge.txt # O*NET knowledge domain ratings (tab-separated)

```
| └─ merged.csv # Final merged dataset (generated by rows.py)
```

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- Scripts (run in this order)

```
| |— rows.py # Step 1: Load, merge, EDA
```

```
| |— model.py # Step 2: Ridge and Lasso regression
```

- | | — clustering.py # Step 3: PCA + K-Means clustering
- | | — classifier.py # Step 4: Logistic regression classifier
- | | — visualizations.py # Step 5: U.S. choropleth map
- |

| — Output Files (generated by scripts)

- | — merged.csv # Generated by rows.py
- | — eda_target_distribution.png # Generated by rows.py
- | — feature_importance.png # Generated by model.py
- | — predicted_vs_actual.png # Generated by model.py
- | — elbow_plot.png # Generated by clustering.py
- | — cluster_scatter.png # Generated by clustering.py
- | — confusion_matrix.png # Generated by classifier.py
- | — classifier_features.png # Generated by classifier.py
- | — us_risk_map.html # Generated by visualizations.py (interactive)
- | — us_risk_map.png # Generated by visualizations.py (static)

Data Sources

All data is publicly available.

Dataset 1 — Frey & Osborne + BLS

- URL:
<https://www.kaggle.com/datasets/andrewmvd/occupation-salary-and-likelihood-of-automation>
- File used: `automation_data_by_state.csv`

- Contains: SOC codes, occupation names, Frey & Osborne automation probability scores (0–1), and employment counts for all 51 U.S. states

Dataset 2 — O*NET Database v29.0

- URL: <https://www.kaggle.com/datasets/emarkhauser/onet-29-0-database>
 - Files used: `Skills.txt`, `Abilities.txt`, `Knowledge.txt`
 - Contains: Numerical skill, ability, and knowledge domain ratings for 900+ U.S. occupations
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Installation

Install all required Python libraries:

```
pip3 install pandas numpy matplotlib seaborn scikit-learn plotly kaleido
```

How to Run

Run the scripts in order. Each script (except `rows.py`) loads `merged.csv`, so `rows.py` must be run first.

Make sure all data files and scripts are in the same directory before running.

Step 1 — Load, merge, and explore the data:

```
python3 rows.py
```

Outputs: `merged.csv`, `eda_target_distribution.png`

Step 2 — Train Ridge and Lasso regression models:

```
python3 model.py
```

Outputs: `feature_importance.png`, `predicted_vs_actual.png`

Step 3 — Run PCA and K-Means clustering:

```
python3 clustering.py
```

Outputs: `elbow_plot.png`, `cluster_scatter.png`

Step 4 — Train logistic regression classifier:

```
python3 classifier.py
```

Outputs: `confusion_matrix.png`, `classifier_features.png`

Step 5 — Generate U.S. choropleth map:

```
python3 visualizations.py
```

Outputs: `us_risk_map.html` (interactive), `us_risk_map.png` (static)

What Each Script Does

`rows.py`

Loads the four source data files, pivots the O*NET files from long to wide format (one row per occupation, one column per skill), merges everything on the shared SOC occupational code, and runs exploratory data analysis. Saves the final merged dataset as `merged.csv` for all subsequent scripts.

- **707 rows × 175 columns** before deduplication
- **120 feature columns** — O*NET skill, ability, and knowledge ratings (all complete, no nulls)
- Target variable mean: **0.506**, median: **0.600**, std: **0.377**
- Distribution is **U-shaped** — spike of safe jobs near 0, rise of high-risk jobs near 1

`model.py`

Trains Ridge and Lasso regression models on the 120 O*NET feature columns to predict the continuous automation probability score (0–1). Uses 80/20 train/test split and cross-validation to select the best regularization strength.

Model	R ²	RMSE	Best Alpha
Ridge	0.657	0.222	100.0
Lasso	0.651	0.224	0.001

Top protective skills (Lasso): Deductive Reasoning, Originality, Fluency of Ideas
Top risk-increasing skills (Lasso): Near Vision, Selective Attention, Inductive Reasoning

clustering.py

Standardizes the 120 feature columns, reduces to 2 dimensions using PCA (57% variance explained), uses the elbow method to select $k=4$, and runs K-Means clustering. Identifies four natural occupation groupings by skill profile.

Cluster	Avg Risk	Count	Job Type
3	0.18	177	Knowledge & Leadership
0	0.54	127	Technical & Trades
1	0.66	132	Analytical & Clerical
2	0.82	181	Physical & Routine Service

classifier.py

Creates a binary high-risk label (Probability > 0.70) and trains a logistic regression classifier. Uses `class_weight='balanced'` to account for slight class imbalance.

Metric	Value
Accuracy	0.831
Precision	0.820
Recall	0.833

Top predictor of high risk: Persuasion
Top predictor of low risk: Fluency of Ideas, Social Perceptiveness

visualizations.py

Calculates a weighted average automation risk score for each U.S. state, weighting each occupation's probability by its employment count in that state. Renders an interactive and static choropleth map.

Highest Risk	Score	Lowest Risk	Score
Nevada	0.665	District of Columbia	0.448
South Dakota	0.657	Massachusetts	0.548
Alabama	0.642	Connecticut	0.568

Key Findings

- O*NET skill profiles explain **65.7% of variance** in automation probability (Ridge $R^2=0.657$)
 - Jobs can be classified as high or low risk with **83.1% accuracy** using skill ratings alone
 - Four distinct occupation clusters emerge naturally from skill profiles, ranging from avg risk 0.18 (leadership roles) to 0.82 (routine physical roles)
 - **Nevada** faces the highest workforce automation exposure; **Washington D.C.** the lowest
 - Skills most protective against automation: Deductive Reasoning, Originality, Social Perceptiveness
 - Skills most associated with high risk: Near Vision, Persuasion, Speed of Limb Movement
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Limitations

- Automation probability scores are sourced from Frey & Osborne (2013) — now over a decade old
- O*NET skill ratings are self-reported and may not perfectly capture actual task composition
- Some creative roles (e.g. Craft Artists, Dancers) appear in the high-risk cluster due to low cognitive complexity ratings, which may not reflect their true automation risk
- The model predicts risk based on occupational skill profiles, not individual worker characteristics